

Multiple Choice

1. **Answer:** B

Options: A) 5; B) 10; C) 2; D) 12

Note: When simplifying $(x\sqrt{x^3})^4$, we first rewrite $\sqrt{x^3}$ as $x^{3/2}$, resulting in $(x \cdot x^{3/2})^4 = (x^{1+3/2})^4 = (x^{5/2})^4$, which simplifies to x^{10} .

2. **Answer:** D

Options: A) 0; B) 1; C) 2; D) 3

Note: The expression simplifies to 2 when $a = 42$, and since $a^{2+1-3} = a^0 = 1$, adding these gives $2 + 1 = 3$.

3. **Answer:** C

Options: A) 252; B) 4680; C) 98770; D) 101170

Note: The correct answer, 98770, is obtained by using the formula for combinations, specifically $nk = \frac{n!}{k!(n-k)!}$, where 8582 simplifies to 853, which calculates to 98770.

4. **Answer:** C

Options: A) $\frac{125}{2916}$; B) $\frac{5}{16}$; C) $\frac{5}{54}$; D) $\frac{5}{55}$

Note: The correct answer is $\frac{5}{54}$ because there are 3 prime numbers (2, 3, 5) and 3 composite numbers (4, 6) on a 6-sided die, and the probability of rolling exactly three primes and three composites can be calculated using combinations and the probabilities of each outcome.

5. **Answer:** D

Options: A) 458; B) 463; C) 547; D) 559

Note: The number 559 cannot be the original number before rounding because it rounds to 600, while numbers between 450 and 549 round to 500.

True / False

6. **Answer:** False

Options: A) True; B) False

Note: The number 179,912 in word form is correctly written as one hundred seventy-nine thousand nine hundred twelve, not one hundred seventy-nine thousand twelve.

7. **Answer:** False

Options: A) True; B) False

Note: The statement is false because the correct number that makes the equation $2/9$ equal to $14/?$ is 63, not 15, as determined by cross-multiplying to find the unknown denominator.

8. **Answer:** False

Options: A) True; B) False

Note: The statement is false because the expression simplifies to $(3 \cdot 2^{20} + 2^{21}) \div 2^{17}$, which equals 10, but the calculation is incorrect because it does not take into account proper simplification and evaluation of powers of 2.

9. **Answer:** False

Options: A) True; B) False

Note: The expression $-1 - 3 \cdot 1 \{ \sim \frac{1}{-3} \sim \}$ simplifies to -3, not 1, because multiplying a negative fraction by the reciprocal of that fraction results in a negative value.

10. **Answer:** True

Options: A) True; B) False

Note: The statement is true because if 1 inch on the map represents 25 miles, then 5 inches would represent 5 times 25 miles, which equals 125 miles.

Long Form Response

11. **Answer:** Note that the numbers increase by 3 each time. Hence we increase by 3 a total of $\frac{32 - (-4)}{3} = 12$ times. But then there must be $12 + 1 = 13$ numbers, since we need to also include the first number on the list.

Note: correct because it determines the total number of terms in the arithmetic sequence by calculating the number of increments of 3 from the starting value of -4 to the endpoint of 32, and then adding one to account for the initial term.

12. **Answer:** The concept of vertical axis of symmetry in vowels refers to the idea that certain vowel shapes can be divided into two mirror-image halves along a vertical line. Most vowels, such as A, O, and U, exhibit this symmetry because they look the same on both sides of the vertical line. However, the vowel E lacks this vertical axis of symmetry, as its shape is not identical when split down the middle. For instance, if you draw a vertical line through the center of E, the left and right sides do not match, demonstrating the absence of symmetry. Therefore, E is the only vowel that does not possess this characteristic.

Note: correct because it accurately identifies the vertical axis of symmetry in vowels, demonstrates that vowels like A, O, and U possess this symmetry, and clearly explains that the vowel E does not have this symmetry due to its asymmetrical shape when divided vertically.

End of Answer Key